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Dapagliflozin and Acute Kidney Injury Following Cardiac Surgery
A Randomized Clinical Trial

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IMPORTANCE Two percent to 50% of patients undergoing elective cardiac surgery experience acute kidney injury (AKI) postoperatively. Medications to prevent AKI after elective cardiac surgery have not been identified.

OBJECTIVE In patients undergoing elective cardiac surgery, to evaluate whether initiating dapagliflozin 1 day prior to surgery reduces the incidence of AKI at 7 days after cardiac surgery, compared with placebo.

DESIGN, SETTING, AND PARTICIPANTS Multicenter, double-blind, placebo-controlled randomized clinical trial conducted at 2 academic medical centers and 5 nonacademic hospitals in the Netherlands. Eligible participants were adults undergoing elective cardiac surgery. Enrollment occurred between June 8, 2023, and January 27, 2025. Final follow-up occurred May 16, 2025.

INTERVENTION Patients were randomized 1:1 to receive either dapagliflozin (10 mg orally; n = 392) or placebo once daily (n = 392), beginning on the day before surgery and continuing through the second postoperative day (total of 4 doses).

MAIN OUTCOME AND MEASURE The primary outcome was the between-group difference in AKI (defined as an increase in serum creatinine level by at least 0.3 mg/dL [26.5 µmol/L] within 48 hours after surgery, a 1.5-fold creatinine increase within 7 days of surgery, or urine output less than 0.5 mL/kg/h for 6 to 12 hours according to Kidney Disease: Improving Global Outcomes criteria) during the first 7 postoperative days.

RESULTS Of 784 participants enrolled, 778 (99%) completed follow-up testing (median age, 68 [61-74] years; 76% male; 97% White; median body mass index, 27 [IQR, 25-30]; and median estimated glomerular filtration rate, 80 [IQR, 67-89] mL/min/1.73 m^2). Compared with placebo, dapagliflozin reduced the incidence of AKI (28% vs 52%; relative risk, 0.54 [95% CI, 0.45-0.65]; $P < .001$) over 7-day follow-up after surgery. Atrial fibrillation and reoperation were the most frequent adverse events. The incidence of atrial fibrillation was 45% (176/392) in the dapagliflozin group vs 45% (176/392) in the placebo group, and the incidence of reoperation was 11% (43/392) vs 10% (39/392), respectively.

CONCLUSIONS AND RELEVANCE In patients undergoing elective cardiac surgery, 4 doses of dapagliflozin, beginning the day before surgery, reduced the incidence of AKI during the 7-day postoperative period.

TRIAL REGISTRATION ClinicalTrials.gov Identifier: NCT05590143

Visual Abstract
Research Summary
Editorial
Supplemental content

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Group Information: A list of the MERCURI-2 Study Group members appears in Supplement 3.

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Two percent to 50% of people experience acute kidney injury (AKI) after cardiac surgery.[1] AKI is associated with development of chronic kidney disease and increased mortality, compared with people without AKI.[2] The pathophysiology of postoperative AKI is complex and multifactorial. Kidney hypoperfusion and hypoxia, ischemia-reperfusion injury, oxidative stress, and inflammation may all contribute to occurrence of AKI after cardiac surgery.[3] To date, no therapies have been identified that reduce the incidence of AKI after cardiac surgery.[3]

Sodium glucose cotransporter 2 (SGLT2) inhibitors were originally developed to lower blood glucose concentrations in people with type 2 diabetes, but SGLT2 inhibitors also reduce the progression of chronic kidney disease.[4] A meta-analysis of 13 clinical trials including 90 413 people with type 2 diabetes and high risk of atherosclerotic cardiovascular disease, heart failure, or chronic kidney disease showed that the SGLT2 inhibitors were associated with lower rates of kidney disease progression (1.97% vs 3.16%)[4] and that SGLT2 inhibitor treatment was associated with lower rates of AKI (1.90% vs 2.60%).[4] SGLT2 inhibitors block the SGLT2 in the proximal tubule, reducing glucose and proximal tubular sodium reabsorption and lowering glomerular pressure, glomerular filtration rate, and kidney workload, while maintaining kidney perfusion.[5] These effects of SGLT2 inhibitors may protect the kidney from hypoxia.[6,7] Other SGLT2 inhibitor–mediated effects that could prevent AKI after cardiac surgery include mitigating ischemia-reperfusion injury, improving left ventricular ejection fraction, and reducing inflammation.[8-10] In an open-label study of 55 patients undergoing cardiac surgery, the incidence of AKI was 20% (5/25) in patients who received empagliflozin vs 66.7% (20/30) in the control group (absolute difference, 46.7% [95% CI, −69.7% to −23.6%]; $P < .001$).[11]

Therefore, the MERCURI-2 (Promoting Effective Renoprotection in Cardiac Surgery Patients by Inhibition of SGLT-2) trial tested the hypothesis that, compared with placebo, dapagliflozin would reduce rates of AKI over the 7-day period following elective cardiac surgery.

Key Points

Question Among patients undergoing elective cardiac surgery, does dapagliflozin, initiated 1 day prior to surgery, reduce acute kidney injury at 7 days after surgery?

Findings In this randomized clinical trial that included 784 patients undergoing elective cardiac surgery, compared with placebo, dapagliflozin significantly reduced the incidence of postoperative acute kidney injury during the 7 days following surgery (52% vs 28%).

Meaning In patients undergoing elective cardiac surgery, these findings support prescribing dapagliflozin to prevent acute kidney injury during the 7-day postoperative period.

Methods

Trial Design and Informed Consent

The MERCURI-2 trial was an investigator-initiated, multicenter, randomized, double-blind, placebo-controlled superiority trial conducted at 7 study sites in the Netherlands. Participating hospitals were located in Amsterdam, Breda, Enschede, Leeuwarden, Leiden, and Zwolle. Enrollment occurred between June 8, 2023, and January 27, 2025. Final follow-up occurred May 16, 2025. The study protocol was approved by the institutional review board of the Amsterdam University Medical Center (METC 2022.0795) and by the competent authorities according to national and international regulations (EudraCT number 2022-002453-25).

All participants provided written informed consent. The protocol was previously published and is available in Supplement 1.[12] The trial was conducted and reported according to the Consolidated Standards of Reporting Trials (CONSORT) guidelines.[13]

Inclusion Criteria

Eligible participants were 18 years or older and scheduled for elective cardiac surgery.

Exclusion Criteria

Exclusion criteria were type 1 diabetes, type 2 diabetes in people with body mass index below 25 (calculated as weight in kilograms divided by square of height in meters) using multiple daily insulin injections (both short-acting and long-acting insulin), baseline estimated glomerular filtration rate (eGFR) below 20 mL/min/1.73 m^2, history of diabetic ketoacidosis, systolic blood pressure below 100 mm Hg at time of evaluation for enrollment, current treatment with SGLT2 inhibitors, and known or suspected allergy to SGLT2 inhibitors. Potential participants who were pregnant, breastfeeding, or intended to become pregnant and women with childbearing potential who were not using adequate contraceptive methods were excluded.

Randomization

Participants were randomized in a double-blind fashion using a 1:1 ratio and random computer-generated block sizes of 4, 6, and 8, stratifying for sex, type 2 diabetes, and study site. Randomization was performed by Castor Electronic Data Capture (Ciwit BV). The randomization results were visible only to the trial pharmacists, who prepared the study medication but were otherwise not involved in the study.

Intervention

Randomized participants received a box with study medication, either dapagliflozin or placebo. To ensure blinding, the original 10-mg dapagliflozin tablets (AstraZeneca) were encapsulated by the Good Manufacturing Practice -certified trial pharmacy at Amsterdam University Medical Center. The capsules were filled to volume with microcrystalline cellulose PH102. The matching placebo capsules contained only microcrystalline cellulose PH102.

In the dapagliflozin group, the first dose of dapagliflozin was administered in the hospital on the day before surgery between 3:00 PM and 8:00 PM; the second dose was administered on the morning of the day of surgery. The third dose was administered on the first postoperative day and the fourth dose on the second postoperative day. Placebo was administered in

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the same manner as the dapagliflozin. Hospital staff and/or study personnel were present during administration of the study medication, and adherence was documented in the electronic case report form. To maintain blinding, the study medication was required to be taken orally and could not be crushed.

Medical History

Participant ethnicity was self-reported from predefined categories. Study personnel recorded results in an electronic case report form. Data on race and ethnicity were collected to provide insight into generalizability of study results. Medical history and medication use were obtained from the medical records.

Primary Outcomes

The primary outcome was the incidence of AKI up to 7 days after cardiac surgery. AKI was defined according to the Kidney Disease: Improving Global Outcomes (KDIGO) criteria[14] as follows: an increase in serum creatinine level of 0.3 mg/dL (26.5 µmol/L) or greater within 48 hours, 1.5-fold or greater increase in creatinine level within 7 days of surgery, or urine output less than 0.5 mL/kg/h for 6 to 12 hours after surgery. For the reference serum creatinine value, the most recent value obtained before initiation of the study medication was used, which typically corresponded to the measurement taken at hospital admission on the day before surgery. To measure the incidence of AKI, daily serum creatinine concentrations were measured the first 7 days after surgery while patients were hospitalized, and hourly urine output was recorded by the clinical team while patients had a urinary catheter in place, typically until they were discharged from the intensive care unit.

Secondary Outcomes

Secondary outcomes included stage 3 AKI (defined below); postoperative maximum change of eGFR within the first 7 days after surgery compared with baseline eGFR; de novo postoperative atrial fibrillation recorded on a 12-lead electrocardiogram; length of stay in the intensive care unit and hospital; incidence of major adverse cardiac events (MACE30), defined as a composite end point of death, new dialysis, or a persistent increase in serum creatinine concentration to 200% or more of the baseline, assessed at 30 days after the operation; incidence of major adverse kidney events (MAKE30), defined as a composite end point of cardiovascular death, nonfatal myocardial infarction, nonfatal ischemic cerebral vascular accident, and hospitalization for heart failure, assessed at 30 days after the operation; World Health Organization Disability Assessment Schedule 2.0 (WHO-DAS 2.0) score within 30 days after the operation (minimally clinical important difference [MCID], 5%)[15]; days at home in the first 30 days (DAH30) after the operation (MCID, 3 days)[16]; and the EuroQol 5 Dimensions 3 Levels (EQ-5D-3L) score measured at 30 days after surgery (MCID, 0.085).[17] Genital mycotic infections, ketoacidosis, and hypoglycemia were included as safety outcomes.

Incidence of MACE30, MAKE30, and safety outcomes were assessed by the research team through medical record review. To measure WHO-DAS 2.0, DAH30, and EQ-5D-3L, patients received a questionnaire 30 days after surgery by email or post. If the patient-reported DAH30 did not match the investigator-calculated DAH30, patients were contacted for verification. The following prespecified secondary end points have not been analyzed at the time of this report and therefore were not reported in this manuscript: health care and productivity costs, perioperative glucose measurements, perioperative hemodynamic vital signs, postoperative cardiac function, and perioperative urinary oxygenation.

Exploratory Outcomes

Exploratory outcomes included the incidence of AKI up to 7 days after cardiac surgery in patients who took all 4 study tablets.

Post Hoc Outcomes

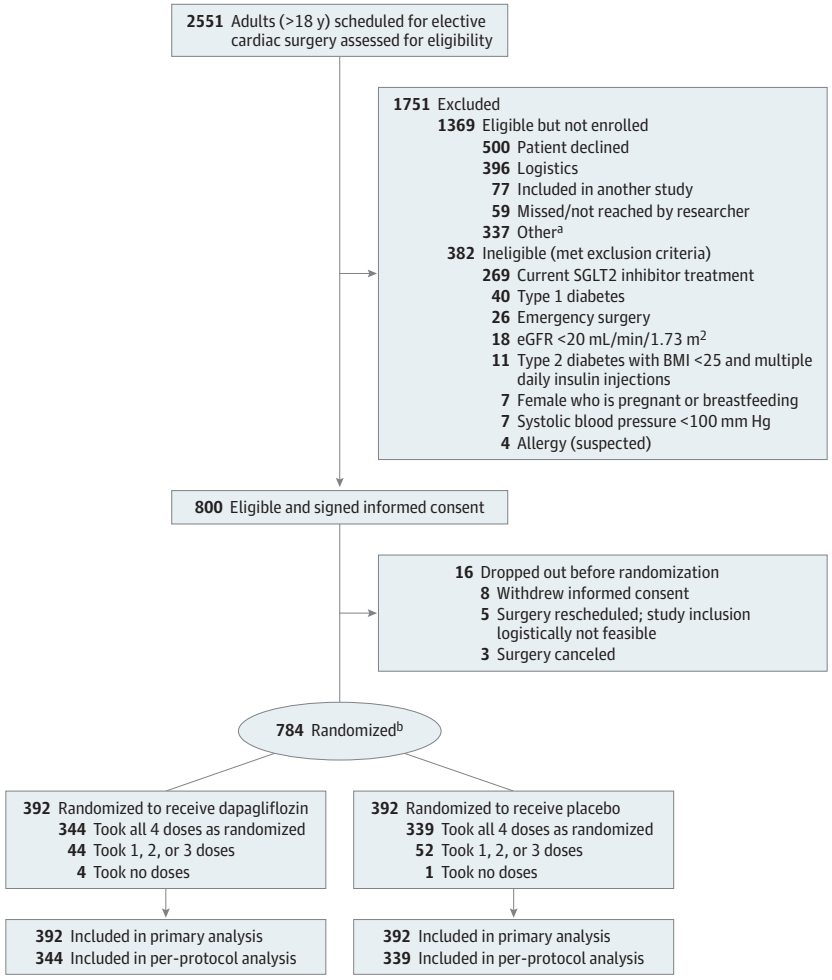
The KDIGO criteria classify AKI into 3 stages. Stage 3 is defined as patients who require kidney replacement therapy, have a serum creatinine increase of 4.0 mg/dL (353.6 µmol/L) or greater or 3 or more times baseline within 7 days, have urine output less than 0.3 mL/kg/h for 24 hours, or experience anuria for more than 12 hours. Stage 2 is classified as a serum creatinine level 2.0 to 2.9 times baseline or urine output less than 0.5 mL/kg/h for more than 12 hours. All other AKI cases are classified as stage 1. Stage 3 AKI was included as secondary end point, stage 1 and 2 as post hoc outcomes. Additional post hoc outcomes included analyses of AKI incidence based on KDIGO criteria, with serum creatinine levels and urine output assessed separately. The following adverse events were post hoc outcomes: incidence of reoperation, atrial arrhythmias, heart failure and/or pleural effusion, pericarditis and/or pericardial effusion, pneumothorax, cerebrovascular stroke or hemorrhage, wound infection, pneumonia, urinary tract infections, and sepsis and/or bacteremia.

Statistical Analysis

Dichotomous data were presented as number with percentage and compared using the χ^2 test or Fisher exact test, as appropriate, and reported as risk differences and relative risks with 95% CIs. Depending on the distribution of the data, continuous variables were reported as the mean and standard deviation or the median and interquartile range and compared using the *t* test or the Mann-Whitney U test. Between-group differences were reported as mean differences with 95% CIs or as estimated median differences calculated using the Hodges-Lehmann estimator with corresponding 95% CIs, depending on the distribution of the data. A Kaplan-Meier curve was generated to illustrate the time patients developed AKI during the first 7 postoperative days and analyzed with the log-rank test. Using a logistic regression model, the primary outcome was adjusted for sex. Heterogeneity of treatment effect between males and females was studied by including a interaction term for group and sex in a logistic regression model. A per-protocol analysis (based on adherence to study medication) for the primary outcome was performed to check the robustness of results. No statistical accounting for missing data was performed. Given the large number of secondary outcomes, the

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Figure 1. Flowchart of the Study Population in the MERCURI-2 Trial



BMI indicates body mass index; eGFR, estimated glomerular filtration rate; SGTL2, sodium-glucose cotransporter 2.

[a]Other reasons included language barrier, not approached for research because the patient did not wish to participate in research in general, not approached because medical team did not approve with study participation.

[b]Randomization stratified for sex, type 2 diabetes, and study site.

secondary outcomes should be considered exploratory. In post hoc analyses, the primary outcome analyses were repeated adjusting for the factors that randomization was stratified on sex, type 2 diabetes, and study site. R version 4.3.2 (R Foundation) was used to calculate estimated median differences, risk differences, and relative risks with corresponding 95% CIs; other statistical analyses were performed with SPSS Statistics version 28 (IBM Corp). $P < .05$ was considered statistically significant.

Sample Size Calculation

Based on previous cohorts, we conservatively estimated an AKI incidence of 22% in the placebo group.[18] SGLT2 inhibitors reduced the risk of AKI, with a relative risk of 0.64 in a previous study.[19] This corresponded to an absolute risk reduction of 7.9%, with an AKI incidence of 14.1% in the intervention group. We calculated that a sample size of 784 would be required to detect this difference, using a 2-sided α of .05 and 80% power. To maintain statistical power, participants who dropped out before randomization were replaced.

Results

Patients

From June 8, 2023, through January 27, 2025, 2551 patients were screened for eligibility. Of these, 800 patients were eligible, and provided written informed consent. A total of 784 patients were randomized, of whom 392 were assigned to dapagliflozin and 392 to placebo (**Figure 1**). Six patients did not undergo surgery; 61 participants did not complete the EQ-5D-3L questionnaire, and 72 participants did not complete the WHO-DAS questionnaire. Characteristics of participants at baseline were similar in the 2 groups (**Table 1**). Participants had a median age of 68 (61-74) years, 76% were male, and 97% were White; overall, participants had a median body mass index of 27 (IQR, 25-30) and median eGFR of 80 (IQR, 67-89) mL/min/1.73 m^2. Twelve percent of participants had type 2 diabetes and 6% had heart failure. Three hundred forty-four patients (88%)in the dapagliflozin group and 339 (87%) patients in the placebo group took all 4 study tablets (eTable 1 in Supplement 2).

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Table 1. Baseline and Surgery Characteristics

	Dapagliflozin (n = 392)	Placebo (n = 392)
Age, median (IQR), y	68 (61-74)	68 (61-74)
Sex, No. (%)		
Female	96 (24)	91 (23)
Male	296 (76)	301 (77)
Ethnic origin, No. (%)		
African	2 (<1)	1 (<1)
Asian	2 (<1)	4 (<1)
Hindustani	3 (<1)	1 (<1)
Moroccan	1 (<1)	3 (<1)
Turkish	3 (<1)	0
White	377 (96)	381 (97)
Other non-Western	1 (<1)	1 (<1)
Other[a]	3 (<1)	1 (<1)
BMI, median (IQR)[b]	27 (25-31)	27 (25-30)
ASA physical status classification, No. (%)[c]		
I (Healthiest)	2 (1)	0
II	29 (7)	22 (6)
III	231 (59)	245 (63)
IV (Least healthy)	130 (33)	125 (32)
NYHA classification, No. (%)[d]		
I	144 (37)	130 (33)
II	135 (35)	161 (41)
III	96 (25)	85 (22)
IV	17 (4)	16 (4)
LVEF, No. (%)		
Good (>50%)	318 (81)	297 (76)
Moderate (31%-50%)	70 (18)	89 (23)
Poor (21%-30%)	4 (1)	6 (2)
EuroSCORE II, median (IQR), %[e]	1.4 (1.0-2.9)	1.6 (1.1-2.8)
Medical history, No. (%)		
Valvular disease	243 (62)	255 (65)
Hypertension	192 (49)	189 (48)
Atrial fibrillation	72 (18)	75 (19)
Current smoking or in the past year	54 (14)	65 (17)
Aneurysm of the thoracic aorta	52 (13)	52 (13)
Chronic obstructive pulmonary disease	49 (13)	48 (12)
Type 2 diabetes	43 (11)	47 (12)
Myocardial infarction in past 90 d	34 (9)	31 (8)
Previous stroke	32 (8)	38 (10)
Peripheral artery disease	28 (7)	39 (10)
Heart failure	25 (6)	21 (5)
Pulmonary hypertension[f]		
Moderate	18 (5)	24 (6)
Severe	5 (1)	7 (2)
Previous cardiac surgery	16 (4)	24 (6)
Atrial flutter	8 (2)	10 (3)
Medication, No. (%)		
Antiplatelet	225 (57)	211 (54)
Statin	222 (57)	227 (58)
β-Blocker	170 (43)	182 (46)
ACE inhibitor	126 (32)	118 (30)
Diuretics	101 (26)	95 (24)
Angiotensin receptor antagonist	77 (20)	79 (20)
DOAC	65 (17)	68 (17)
Metformin	32 (8)	31 (8)
Sulfonylurea derivate	14 (4)	18 (5)
Vitamin K antagonist	9 (2)	15 (4)
Insulin treatment	4 (1)	6 (2)
eGFR, median (IQR), mL/min/1.73 m^2[g]	79 (66-89)	80 (67-89)
Details of surgery		
Type of surgery, No./total (%)[h]		
≥2 Procedures	156/388 (40)	165/390 (42)
Single non-CABG procedure	110/388 (28)	121/390 (31)
CABG procedure	122/388 (31)	104/390 (27)
Duration of surgery, median (IQR), min	204 (167-259)	206 (168-264)
Cardiopulmonary bypass assisted, No./total (%)	348/388 (90)	349/ 390 (90)
Duration cardiopulmonary bypass, median (IQR), min	110 (80-153)	108 (80-150)
Clamp duration, median (IQR), min	76 (55-104)	76 (54-108)

Abbreviations: ACE, angiotensin-converting enzyme; ASA, American Society of Anesthesiologists; BMI, body mass index; CABG, coronary artery bypass graft; DOAC, direct oral anticoagulant; eGFR, estimated glomerular filtration rate; EuroSCORE, European System for Cardiac Operative Risk Evaluation; LVEF, left ventricular ejection fraction; NYHA, New York Heart Association.

[a] Reported because patients either did not know their background or were of mixed descent.

[b] Calculated as weight in kilograms divided by square of height in meters.

[c] A physician-assessed preoperative score that ranges from I (healthy patient with no systemic disease) to V (moribund) and VI (brain death).

[d] NYHA I indicates no symptoms and no limitation in ordinary physical activity; NYHA II, slight limitation during ordinary activity; NYHA III, marked limitation in activity; NYHA IV, severe limitations, symptoms even while at rest.

[e] A validated multivariable model including patient-, cardiac-, and surgery-related factors; range, 0% to 100%, with higher values indicating higher predicted risk of in-hospital mortality after cardiac surgery.

[f] Moderate pulmonary hypertension indicates pulmonary artery systolic pressure 30-55 mm Hg; severe pulmonary hypertension, pulmonary artery systolic pressure 55 mm Hg or greater.

[g] Calculated with the CKD-EPI (Chronic Kidney Disease Epidemiology Collaboration) formula; normal eGFR is 90 mL/min/1.73 m^2 or greater.

[h] Procedures included CABG surgery, valve repair or replacement, or aortic root or arch repair or replacement.

Four patients in the dapagliflozin group and 1 patient in the placebo group took no study tablets.

Primary Outcome

AKI occurred in 111 participants randomized to the dapagliflozin group (28%) and 205 randomized to placebo (52%)

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Table 2. Primary Outcome, Components of the Primary Outcome, and Secondary Outcomes

Outcome	Dapagliflozin (n = 392)	Placebo (n = 392)	Unadjusted		*P* value
			Between-group difference in risk or median	Relative risk (95% CI)	
Primary outcome					
Acute kidney injury, No. (%)	111 (28)	205 (52)	−23.98 (−30.64 to −17.32)	0.54 (0.45 to 0.65)	<.001
Secondary outcomes					
KDIGO stage 3 (kidney replacement therapy, creatinine ≥353.6 µmol/L or ≥3.0 × baseline, urine output <0.3 mL/kg/h for >24 h or anuria for >12 h), No. (%)	3 (0.8)	1 (0.3)	0.51 (−0.49 to 1.51)	3.00 (0.31 to 28.71)	.62
Postoperative maximum change in eGFR vs baseline, median (IQR), mL/min/1.73 m^2	−7.4 (−14.8 to −1.67)	−1.75 (−11.2 to 2.63)	4.39 (2.98 to 5.82)		<.001
De novo atrial fibrillation, No. (%)	127 (32)	121 (31)	1.53 (−4.98 to 8.04)	1.05 (0.85 to 1.29)	.70
MACE30, No. (%)[a]	22 (5.6)	24 (6.1)	−0.51 (−3.80 to 2.78)	0.92 (0.52 to 1.61)	.88
MAKE30, No. (%)[b]	5 (1.3)	2 (0.5)	0.77 (−0.55 to 2.08)	2.50 (0.49 to 12.81)	.45
Duration of length of stay in ICU, median (IQR), d	1 (1-1)	1 (1-1)	0 (0 to 0)[c]		.39
Duration of stay in hospital, median (IQR), d	6 (5-9)	6 (5-9)	0 (0 to 0)[c]		.85
DAH30, median (IQR), d[d]	23 (20-25)	23 (19-25)	0 (0 to 0)[c]		.69
Quality of life at 30 d, median (IQR) EQ-5D-3L score[e]	0.76 (0.69-0.88) [n = 362]	0.76 (0.69-0.88) [n = 361]	0 (0 to 0)[c]		.82
Functioning at 30 d, median (IQR) WHO-DAS score[f]	13 (6-23) [n = 354]	13 (6-25) [n = 358]	0 (−2.08 to 2.08)[c]		.65
Post hoc outcomes					
KDIGO stage 1 (creatinine ≥26.5 µmol/L or ≥1.5 × baseline, urine output <0.5 mL/kg/h for 6-12 h), No. (%)	92 (23)	155 (40)	−16.07 (−22.48 to −9.67)	0.59 (0.48 to 0.74)	<.001
KDIGO stage 2 (creatinine 2.0-2.9 × baseline, urine output <0.5 mL/kg/h for >12 h), No. (%)	16 (4.1)	49 (13)	−8.42 (−12.23 to −4.60)	0.33 (0.19 to 0.56)	<.001

Abbreviations: DAH30, days at home in first 30 days; eGFR, estimated glomerular filtration rate; EQ-5D-3L, EuroQol 5 Dimensions 3 Levels; ICU, intensive care unit; KDIGO, Kidney Disease: Improving Global Outcomes; MACE, major adverse cardiac events; MAKE, major adverse kidney events; WHO-DAS, World Health Organization Disability Assessment Schedule.

SI conversion factor: To convert creatinine values to md/dL, divide by 88.4.

[a] MACE30 is number of patients with cardiovascular death, a new myocardial infarction, a cerebrovascular accident confirmed by CT-scan, or hospital readmission due to heart failure within 30 days.

[b] MAKE30 is number of patients with kidney related death, with kidney failure requiring renal replacement therapy or persistent kidney failure, defined as serum creatinine concentrations ≥200% of the baseline value 30 days after surgery.

[c] Data show the estimated median difference with 95% CI.

[d] DAH30: Calculated as the number of days a patient was alive and not hospitalized during the first 30 days after surgery. Range: 0 to 30 days, with higher values indicating better outcomes (more days alive and out of hospital).

[e] A standardized instrument measuring health-related quality of life across 5 domains (mobility, self-care, usual activities, pain/discomfort, anxiety/depression). Index scores range from 0 (worst health) to 1 (perfect health), with higher scores indicating better quality of life.

[f] WHO-DAS assesses disability across six domains of functioning (cognition, mobility, self-care, getting along, life activities, participation). Scores are computed as a percentage, ranging from 0% (no disability) to 100% (full disability), with higher scores indicating greater disability.

(risk difference, −23.98% [95% CI, −30.64% to −17.32%]; relative risk [RR], 0.54 [95% CI, 0.45-0.65]; *P* < .001) (**Table 2** and **Figure 2**).

Secondary Outcomes

Three patients in the dapagliflozin group (0.8%) and 1 patient in the placebo group (0.3%) had stage 3 AKI (RR, 3.0 [95% CI, 0.31-28.71]; *P* = .62). Postoperative maximum change in eGFR compared with baseline was −7.4 (95% CI, −14.8 to −1.67) mL/min/1.73 m^2 in the dapagliflozin group and −1.75 (95% CI, −11.2 to 2.63) mL/min/1.73 m^2 in the placebo group (median difference, 4.39 [95% CI, 2.98-5.82]; *P* < .001) (for creatine measurements see eFigure in Supplement 2). There were no between-group differences in the incidence of de novo atrial fibrillation (32% vs 31%; RR, 1.05 [95% CI, 0.85-1.29]), MACE30 (5.6% vs 6.1%; RR, 0.92 [95% CI, 0.52-1.61]), MAKE30 (1.3% vs 0.5%; RR, 2.50 [95% CI, 0.49-12.81]), duration of hospital stay (median, 6 [IQR, 5-9] vs 6 [IQR, 5-9] days; median difference, 0 [95% CI, 0-0]), or DAH30 (median, 23 [IQR, 20-25] vs 23 [IQR, 19-25]); median difference, 0 [95% CI, 0-0]).

After adjustment for sex in a multivariable logistic regression model, dapagliflozin remained associated with a lower incidence of AKI (odds ratio, 0.36 [95% CI, 0.27-0.48]; *P* < .001). Subgroup analyses based on sex showed no heterogeneity in treatment (*P* = .89).

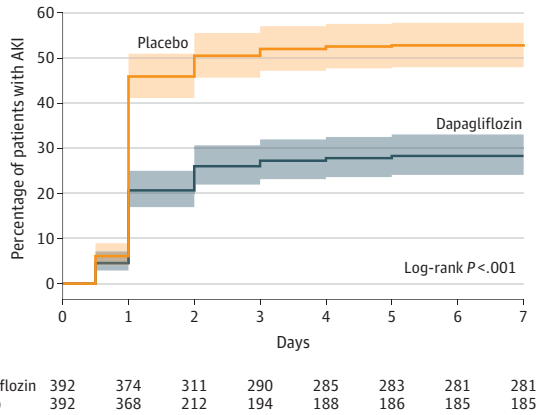
Exploratory Outcomes

In exploratory analyses, the primary outcome was analyzed using a per-protocol analysis method in which only patients who took all 4 tablets of study medication were included (eTable 2 in Supplement 2). In the per-protocol analysis, the incidence of AKI was 27% (92/344) in the dapagliflozin group

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Figure 2. Kaplan-Meier Plot of Time to Acute Kidney Injury in the Primary Analysis



No. at risk

Dapagliflozin	392	374	311	290	285	283	281	281
Placebo	392	368	212	194	188	186	185	185

Proportion of patients with acute kidney injury (AKI; primary outcome) within 7 days after surgery. Shaded regions indicate 95% CIs.

and 51% (173/339) in the placebo group (relative risk, 0.52 [95% CI, 0.43-0.64]).

Post Hoc Outcomes

Stage 1 AKI occurred in 92 participants (23%) in the dapagliflozin group and 155 (40%) in the placebo group. Stage 2 AKI occurred in 16 patients (4%) in the dapagliflozin group and 49 (13%) in the placebo group. When AKI was defined exclusively by creatinine criteria, AKI occurred in 53 (14%) in the dapagliflozin group and 57 (15%) in the placebo group (RR, 0.93 [95% CI, 0.66-1.32]) (eTable 3 in Supplement 2). When AKI was defined exclusively by urine output criteria, AKI incidence was 21% (82/392) in the dapagliflozin group and 48% (188/392) in the placebo group (RR, 0.44 [95% CI, 0.35-0.54]). After adjustment for the stratification factors sex, type 2 diabetes, and site, the adjusted event rate was 28.3 (95% CI, 23.9-32.7) in the dapagliflozin group and 52.3 (95% CI, 47.4-57.2) in the placebo group.

Adverse Events

Genital mycotic infections occurred in 1 participant (0.26%) in the dapagliflozin group and 0 in the placebo group; ketoacidosis in 1 (0.26%) in the dapagliflozin group and 0 in the placebo group; and hypoglycemia in 1 (0.26%) in the dapagliflozin group vs 0 in the placebo group (**Table 3**). The most common post hoc adverse events were atrial fibrillation (45% vs 45%) and reoperation (11% vs 10%).

Table 3. Postoperative Adverse Events

Event	No. (%) Dapagliflozin (n = 392)	Placebo (n = 392)
Atrial fibrillation	176 (45)	176 (45)
Reoperation	43 (11)	39 (10)
Heart failure and/or pleural effusion	40 (10)	41 (10)
Pericarditis and/or pericardial effusion	38 (10)	25 (6)
Pneumonia	22 (6)	29 (7)
Atrial flutter	22 (6)	13 (3)
Wound infection	14 (4)	16 (4)
Pneumothorax	8 (2)	6 (2)
Cerebrovascular stroke or hemorrhage	7 (2)	3 (1)
Urinary tract infection	6 (2)	13 (3)
Sepsis and/ or bacteremia	3 (1)	4 (1)
Hypoglycemia	1 (0.26)	0
Ketoacidosis	1 (0.26)	0
Genital mycotic infection	1 (0.26)	0

Discussion

Among adult patients undergoing elective cardiac surgery, perioperative treatment with dapagliflozin reduced the incidence of postoperative AKI compared with placebo. However, dapagliflozin did not improve any secondary outcomes, including MAKE30, de novo atrial fibrillation, length of hospital stay, or any other secondary outcomes.

These results were consistent with those of the pilot study, in which perioperative empagliflozin was associated with a lower incidence of AKI after cardiac surgery.[11] In addition, a recent randomized, open-label study in patients undergoing cardiac surgery showed that empagliflozin reduced the incidence of postoperative AKI (AKI incidence, 22.5% [16/71] in the empagliflozin group vs 39.1% [29/74] in the control group; absolute difference, −16.7% [95% CI, −31.4% to −1.9%]; $P = .03$).[20]

In the current study, AKI incidence was 52% in the placebo group. Prior literature showed that rates of AKI incidence after cardiac surgery ranged from 2% to 50%. The relatively higher AKI incidence rate in the placebo group in the current trial is likely due to the inclusion of urine output less than 0.5 mL/kg/h for 6 to 12 hours as a criterion for the primary outcome. Of participants who experienced the primary outcome in this trial, 48% in the placebo group and 21% in the dapagliflozin group met the primary outcome criterion based on reduced urine output. The relatively high incidence of AKI observed in this study was in part due to reductions in urine outcome after surgery. AKI defined solely on the basis of urine output was also associated with adverse long-term outcomes, including an increased risk of MAKE events at 180 days and 1-year mortality.[21,22] Despite initial suggestions, SGLT2 inhibitors do not seem to have diuretic actions; carefully designed studies with fixed sodium intake and 24-hour urine collections found no significant increase in urine production.[5,23-28] Only the study by Heise et al[27] found an increase in urine volume after the first dose of empagliflozin, which returned to baseline after multiple doses. In studies reporting increased diuresis, this effect was limited to the first day of treatment,[29,30] and the increase in urine output did not exceed approximately 400 mL.[31]

The strengths of this trial included its large, randomized, placebo-controlled, multicenter design and evaluation of an intervention that could be easily implemented in clinical practice.

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Limitations

This trial had several limitations. First, the primary outcome was a 7-day outcome and there were no statistically significant between-group differences in secondary outcomes. Second, the study did not have statistical power to detect differences in stage 3 AKI or MAKE30 (composite of death, new dialysis, and worsened kidney function) outcomes. Third, no histological samples or biomarkers were analyzed to compare the extent of kidney injury between the 2 treatment groups. Fourth, specific treatment protocols for AKI prevention were not implemented. The absence of study-specific protocols enhances the generalizability of the findings to routine clinical practice. Fifth, the population included 97% White individuals who were not already using SGLT2 inhibitors (an exclusion criterion), which may have limited the number of participants with type 2 diabetes, heart failure, and chronic kidney disease, potentially limiting the generalizability of results. Sixth, 76% of participants were male. Seventh, results were not generalizable to patients not undergoing elective cardiac surgery. Eighth, no adjustments were made for multiple comparisons of the secondary outcomes. Ninth, the prespecified analyses did not include adjustment for stratification factors, which may have introduced bias. Therefore, post hoc analyses of the primary outcomes were repeated, adjusting for the randomization stratification (sex, type 2 diabetes, and site). Tenth, findings may not be generalizable to other SGLT2 inhibitors. Eleventh, data were not collected on the number (proportion) of participants without creatinine measurement on the day before surgery, so the time between baseline measurement of serum creatinine level and surgery could not be analyzed.

Conclusions

In patients undergoing elective cardiac surgery, 4 doses of dapagliflozin, beginning the day before surgery, reduced the incidence of AKI during the 7-day postoperative period.

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